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Block rotations along the strike-slip Finlay-Ingenika fault, north-central British Columbia: Implications for paleomagnetic and tectonic studies

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Abstract. Structure developed in association with oblique motions on the dextral, strike-slip Finlay-Ingenika fault (FIF) in the McConnell Creek area, north-central British Columbia, is dominated by subvertical to vertical strike-slip faults. Some of the faults cut the area into discrete, fault-bounded blocks several kilometres wide. Stress tensors for 24 sites were inverted from the regional cleavage which predates the block-bounding faults. The site-mean stress tensors indicate that the fault-bounded blocks were rotated clockwise about a subvertical axis during progressive oblique motions on the FIF. Rotation varies systematically, being maximal ($58.7\pm3.3°$) close to the FIF and minimal ($0.0\pm1.6°$) about 20 km away. Paleomagnetic samples were collected from 13 sites in the widespread Early Jurassic to Cretaceous plutonic rocks. Interpretable magnetic components of presumed Late Cretaceous age were obtained from six sites. The observed paleopole is significantly different from the Late Cretaceous reference pole for cratonal North America (CNA). After corrections for the local block rotations, the precision is much improved (K increased from 48 to 383), and the paleopole moves closer to the reference pole but is not coincident with it. Rotation about the Eulerian pole for the best-fitting small circle to the Tintina trench and northern Rocky Mountain trench fault zone, however, brings the observed pole into coincidence with that for CNA and requires ~670 km of dextral displacement on the fault zone. It is evident therefore that local structures associated with large dextral strike-slip faults could account for at least part of the paleomagnetic disparity between some western parts of the Canadian Cordillera and CNA.

Introduction

Cretaceous rocks of the western Cordillera always yield discordant paleomagnetic directions (see reviews by *Beck* [1992a] and *Irving and Wynne* [1992]). Many data are from plutonic rocks whose tilts were unknown at the time the observations were made so that magnetization directions could not be confidently restored to their original orientation. Recently, petrological evidence (bathozonal mineral assemblage in contact rocks [*Irving and Archibald*, 1990; *Brown and Burmester*, 1991] and aluminium-in-hornblende bathozonal studies [*Ague and Brandon*, 1992]) have been used to estimate tilts of some plutons, allowing the magnetization directions to be corrected to paleohorizontal plane at the time of magnetization acquisition. Another procedure, proposed here, is to use detailed structural observations from the plutons and neighbouring rocks to restore the magnetization to their original orientation. The method is illustrated from the eastern Intermontane Belt, the western Canadian Cordillera. The area is structurally very complex, and the rocks often do not preserve the paleomagnetic field so results are few. They are, however, sufficient to illustrate the viability of the method and provide useful constraints on tectonic motions.

The Cretaceous discordant magnetization directions have previously been interpreted as results of tilting [cf. *Beck and Noson*, 1972; *Beck et al.*, 1981; *Butler et al.*, 1989, 1990; *Irving and Wynne*, 1990], northward transport and clockwise rotation [cf. *Beck and Noson*, 1972; *Monger and Irving*, 1980; *Beck et al.*, 1981; *Irving et al.*, 1985; *Umhoefer*, 1987; *Irving and Wynne*, 1990; *Umhoefer and Magloughlin*, 1990; *Irving and Wynne*, 1991; *Wynne et al.*, 1995], or combination of tilting and northward transport [cf. *Irving and Wynne*, 1990; *Umhoefer and Magloughlin*, 1990]. Both tilting and clockwise rotation about vertical axes should give rise to structural features observed in the field unless the entire western Canadian Cordillera experienced them as one block, which seems unlikely [*Beck*, 1992b].

We have carried out a detailed structural analysis of part of the eastern Intermontane Belt of north-central British Columbia adjacent to the dextral strike-slip Finlay-Ingenika fault (FIF), from which aberrant paleomagnetic poles had been reported [e.g., *Monger and Irving*, 1980; *Wynne et al.*, 1992]. Kinematic indicators, rock and magnetic fabrics, as well as strain measurements along the faults in the study area and their geometric relationships have previously been described and discussed by *Zhang and Hynes* [1994, 1995]. We will not present these data here so as to avoid significant

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duplication, and for details of the structural analysis, readers are referred to *Zhang and Hynes* [1994, 1995]. In this paper, we review the structural characteristics of the region and describe the paleomagnetism of the plutonic rocks there. Finally, we describe how structural analysis may be used to correct discordant paleomagnetic directions.

Tectonic Setting and Major Structures

The study area lies within the eastern part of the Intermontane Belt in north-central British Columbia (Figure 1) and straddles the boundary between the Quesnellia and Stikinia tectonostratigraphic terranes [*Monger*, 1984]. Stikinia, Cache Creek, Quesnellia, and Slide Mountain terranes had been essentially amalgamated by the Early Jurassic due to the closing of the Cache Creek ocean, forming the composite Terrane I, and were subsequently accreted to the ancient margin of North America during Middle Jurassic contraction [*Monger*, 1984]. Transcurrent motions between the Intermontane Belt and North American craton occurred in the Late Cretaceous to Eocene at a time when the Kula plate in the west was converging obliquely with the North American plate [*Engebretson et al.*, 1985; *Umhoefer*, 1987]. These motions apparently have produced dextral strike-slip displacements along the Tintina trench and northern Rocky Mountain trench (TT-NRMT) fault zone [*Roddick*, 1967; *Tempelman-Kluit*, 1979; *Gabrielse*, 1985; *Struik*, 1993] and a sequence of faults

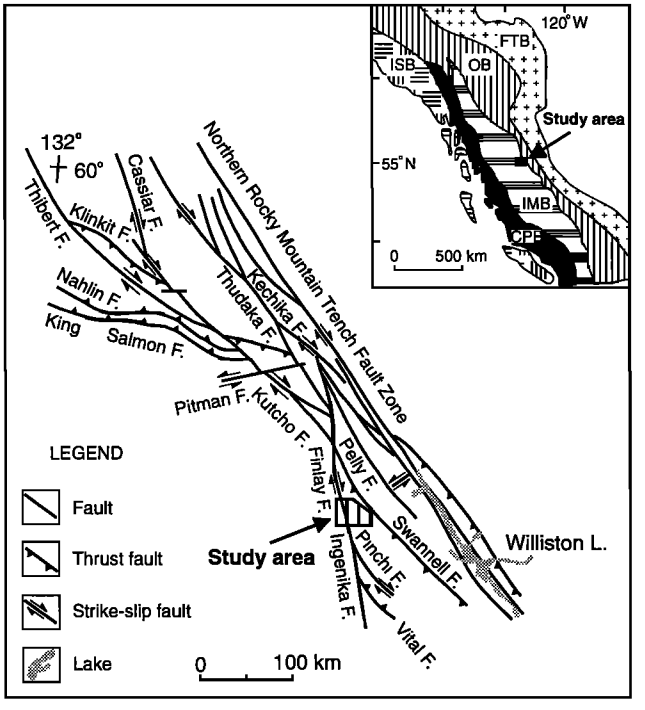


Figure 1. Generalized structural map of north-central British Columbia [after *Gabrielse*, 1985] showing the tectonic setting and location of the study area. Inset shows the main tectonic units of the Canadian Cordillera. Abbreviations are FTB, Fold and Thrust Belt; OB, Omineca Belt; IMB, Intermontane Belt; CPB, Coast Plutonic Belt; ISB, Insular Belt [after *Wheeler and McFeely*, 1991].

further west (Figure 1). *Gabrielse* [1985] has shown that about 110 km of dextral displacement along the FIF may be required for the restoration of Kutcho fault in the north with Pinchi fault in the south (Figure 1).

The area is underlain by Late Triassic Takla Group volcaniclastic and minor sedimentary rocks [*Monger*, 1977], which are extensively intruded by Early Jurassic to Cretaceous plutonic rocks [*Lord*, 1948; *Roots*, 1954; *Richards*, 1976a, b; *Woodsworth*, 1976]. The structural analysis [*Zhang and Hynes*, 1994, 1995] indicates that displacement along the FIF is dextrally transpressive. Structure resulting from the dextrally transpressive displacement on the FIF is dominated by subvertical to vertical, dextral strike-slip faults trending northwest, north-northwest, and north-south and sinistral strike-slip faults trending east-northeast (Figure 2), as well as some large-scale northwest to north-northwest trending folds. Based on their geometric and spatial relationships, all these faults can be related to the stress field associated with movement on the FIF [*Zhang and Hynes*, 1994]. Strain analysis of the deformed volcanic fragments along one of the north-northwest trending, dextral strike-slip faults shows that the principal orientations of the strain ellipsoids are in accordance with those of the magnetic susceptibility ellipsoids from the adjacent deformed dioritic rocks, indicating that the plutonic rocks experienced the same deformation as the Late Triassic Takla Group volcanics [*Zhang and Hynes*, 1994].

The dextral displacement of ~110 km along the FIF probably took place during middle Cretaceous or Late Cretaceous time, based on its crosscutting relationship with the Kutcho and Pinchi faults [*Gabrielse*, 1985]. The strike-slip faulting in the study area may therefore not have been earlier than middle Cretaceous as Early Jurassic to Cretaceous dioritic rocks were involved in the faulting in many places. Some of the major strike-slip faults cut the Late Triassic Takla Group rocks into a number of discrete blocks with diameters ranging from a few to tens of kilometres. Internally, the blocks are only weakly deformed. As dextral transpressive displacement on the FIF progressed, deformation was apparently concentrated in the previously formed fault zones, while the fault-bounded blocks remained relatively undeformed. Within the fault-bounded blocks, the only visible deformation is a spaced cleavage. This cleavage has permitted us to track the relative motions of adjacent fault blocks.

Statistics of Regional Cleavage

The cleavage within the fault-bounded blocks has been classified into local and regional (see *Zhang and Hynes* [1994] for details). The locally distributed cleavage is produced by displacement on the block-bounding strike-slip faults and is restricted to their vicinity. The regionally distributed cleavage is locally truncated and rotated by nearby block-bounding faults and hence predates the extensive strike-slip faulting in the study area. On the other hand, since the regional cleavage surfaces show consistently steep dips throughout the study area (Figure 3a), they postdate the folding. The regional cleavage was therefore probably produced during the early stage of dextral displacement on

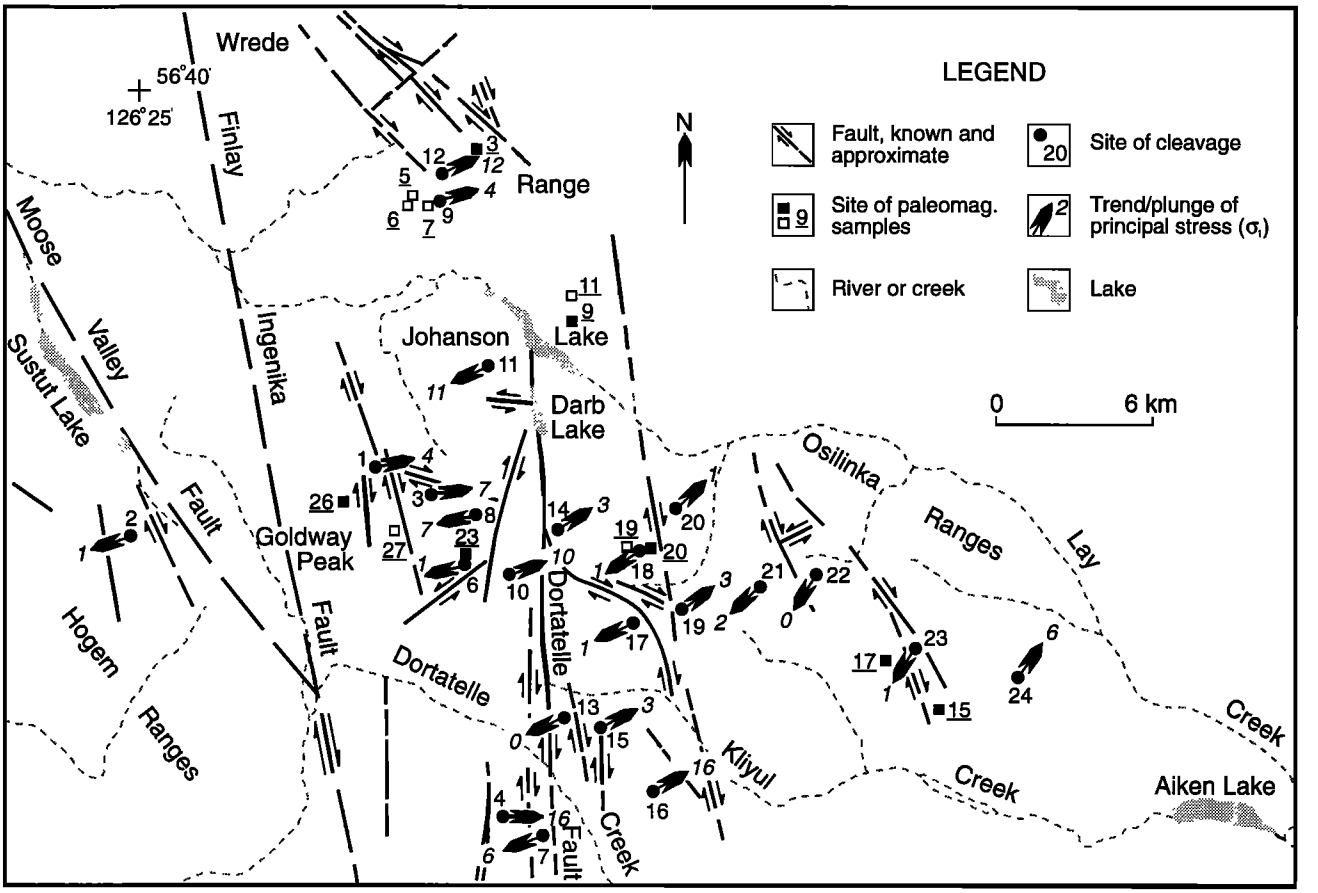


Figure 2. Generalized structural map of the Johanson Lake area [after *Zhang and Hynes*, 1994]. Cleavage site 5 is from west of McConnell Creek and outside the map sheet. Open squares show the site of interpretable paleomagnetic samples.

the FIF based on its geometric and spatial relationships to the FIF [*Zhang and Hynes*, 1994]. This cleavage sometimes occurs in conjugate sets and has slickenlines on its surfaces.

About 600 regional cleavage surfaces, together with slicken lines, were measured from 24 sites (Figure 2). Subhorizontal slickenlines on the cleavage surfaces (Figure 3b), together with steeply plunging conjugate cleavage-surface intersections, indicate that the cleavage was produced by a stress state with subvertical intermediate principal stress (σ_2). Strikes of the cleavage surfaces, however, vary greatly over the area (Figure 3a). This variation is particularly evident from conjugate cleavages. For example, close to the FIF, conjugate cleavage sets strike northeast and east-southeast (Figure 4a), about 12 km from the fault they have changed their strikes to north-northeast and east-west (Figure 4b), and at about 18 km they are approximately north-south and east-northeast (Figures 4c and 4d).

Since the conjugate cleavages within a single fault-bounded block show the same attitude (e.g., Figures 4c and 4d), the variation in strikes of the cleavage surfaces is probably due to differential block rotation rather than gradual change in the trajectories of cleavage through the area. On this assumption, the bisector of, for example, the acute dihedral angle between conjugate cleavage sets could be used as a monitor of the rotation between blocks. There are, however, several blocks for which both cleavage sets are not well developed; instead, there is just one set of the conjugate cleavage, with associated slickenlines. In order to use all our cleavage data, we have inverted the cleavage and slickenline data to determine possible configurations of the stress tensor for each site and have then used the statistical distributions of possible stress tensors to monitor changes between blocks. We emphasize that the overall character of our conclusions is not affected by the use of inverted stress results, rather than the strain data themselves, but the precision is markedly improved, because we have so many more data to use. Furthermore, with this increased precision we are able to strengthen the evidence for our inference that cleavage attitudes are consistent within but not between blocks.

Inversion of Stress Tensors from Fault-Slip Data

The principle of inversion of stress tensor from fault-slip data is straightforward [e.g., *Angelier*, 1984; *Michael*, 1984; *Reches*, 1987; *Hardcastle and Hills*, 1991]. In terms of classical solid mechanics, given a stress state **S** (stress tensor), the total stress **p** on any given plane can be calculated based on the force-balance equations

$$\mathbf{p} = \begin{pmatrix} \sigma_{xx} & \tau_{yx} & \tau_{zx} \\ \tau_{xy} & \sigma_{yy} & \tau_{zy} \\ \tau_{xz} & \tau_{yz} & \sigma_{zz} \end{pmatrix} \begin{pmatrix} l \\ m \\ n \end{pmatrix} = \mathbf{S}\cdot\mathbf{n} \qquad (1)$$

where **S** is a symmetric tensor and **n** is a unit vector normal to the plane. The resolved normal and shear stresses on the plane are

$$\sigma = \mathbf{n}^T\cdot\mathbf{S}\cdot\mathbf{n}\cdot\mathbf{n} \qquad (2a)$$

and

$$\tau = \mathbf{p}-\sigma = \mathbf{S}\cdot\mathbf{n}-\mathbf{n}^T\cdot\mathbf{S}\cdot\mathbf{n}\cdot\mathbf{n} \qquad (2b)$$

respectively, where $\mathbf{n}^T$ is the transpose of **n**. In our analysis, a stress tensor is considered a candidate for the one producing the observed cleavage and slickenline at a sample site, if (1) the resolved shear stress satisfies the Mohr-Coloumb yield condition $\tau \geq C+\mu\sigma$, where $\tau = |\tau|$, $\sigma = |\sigma|$, and C and μ are the cohesion and coefficient of friction of the rock mass under study, respectively; (2) the angle between the resolved shear stress (τ) in (2b) and the slickenline observed in the field is less than 25°; and (3) more than 80 percent of the faults collected from the site satisfy conditions (1) and (2).

This inversion technique assumes that the stress state is uniform and that the rock mechanical properties (i.e., C and μ) are homogeneous and isotropic, within the area of data collection [*Angelier*, 1984; *Michael*, 1984; *Reches*, 1987; *Hardcastle and Hills*, 1991]. The choice of μ is based on both experimental and field observations. In rock mechanical experiments, *Byerlee* [1978] demonstrated that $\mu \approx 0.85$ at intermediate pressures. In the presence of clay minerals, however, faults yield low values of μ [*Radney and Byerlee*, 1988]. On the other hand, natural faults often show very low strength; for example, some faults in the San Andreas fault system exhibit shear strength a factor of 2 to 3 lower than predicted by frictional theory [*Zoback et al.*, 1987]. One reason for this is probably that the faults possess low values of μ due to the presence of clay minerals, which is in accord with other field observations of low values of μ (e.g., $\mu <$ 0.1 to $\mu = 0.8$ [cf. *Reches*, 1987]). Since the slickensides observed in our study area are almost all composed of hydrous minerals such as chlorite, epidote, and tremolite, we selected a value of $\mu = 0.3$, which is consistent with that suggested by *Hardcastle and Hills* [1991]. Because the observed slickenlines recorded the motions on the fault surfaces after their rupture, we chose $C = 0.0$. This simplification affects only the size of the deviatoric stress but not principal directions of the stress tensor and has therefore no influence on our conclusions.

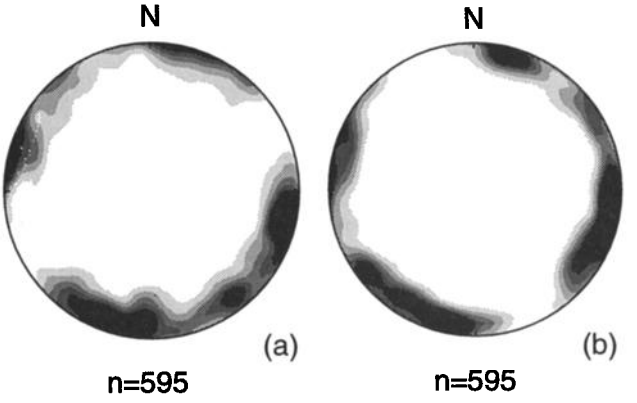


Figure 3. Stereographic plots of poles to (a) cleavage surfaces and (b) slickenlines. Contouring in this and subsequent figures uses Gaussian weighting function [cf. *Robin and Jowett*, 1986]. The lowest contour level is $E = 3\sigma$, and unless otherwise noted, contours are in increments of 2σ thereafter, where σ is the statistical dispersion of the data. Here n is the number of measurements. All stereograms in this paper are lower hemisphere, equal-area projections except those for paleopoles which are upper hemisphere, equal-area projections.

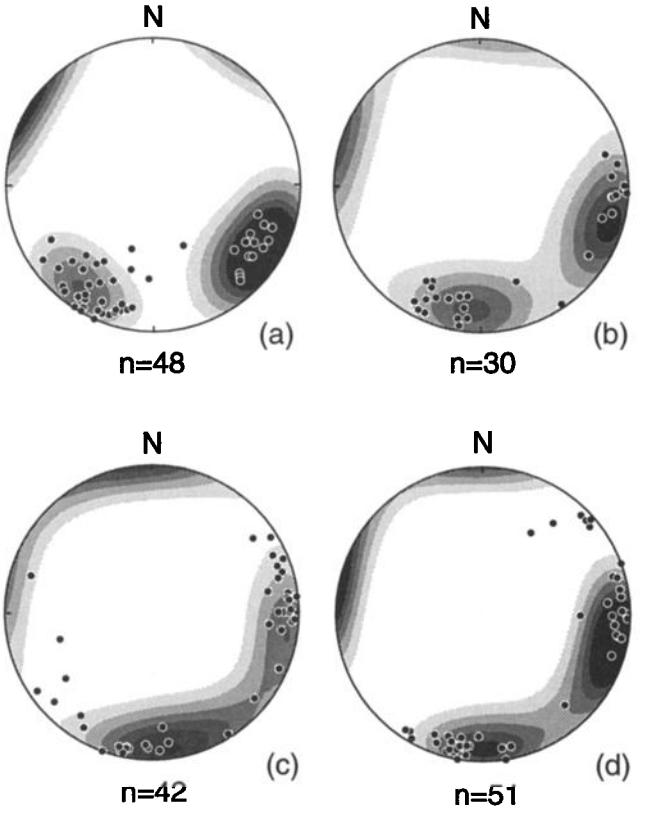


Figure 4. Contours of poles to conjugate cleavage surfaces from sites (a) 1, (b) 18, (c) 20, and (d) 21 (see Figure 2 for locations of sites). Contour increment is 2σ; solid circles are slickenlines; n is the number of measurements.

Stress tensors with principal axes at intervals of 10° in space were tested against the data from each site using the program BRUTE3 [*Hardcastle and Hills*, 1991]. A few hundred to thousands of possible stress tensors were obtained for each site.

Determination of Mean Directions of Stress Tensors and Their 95% Confidence Limits

Despite the range in distributions of the inverted stress tensors, they generally exhibit strongly defined maxima for

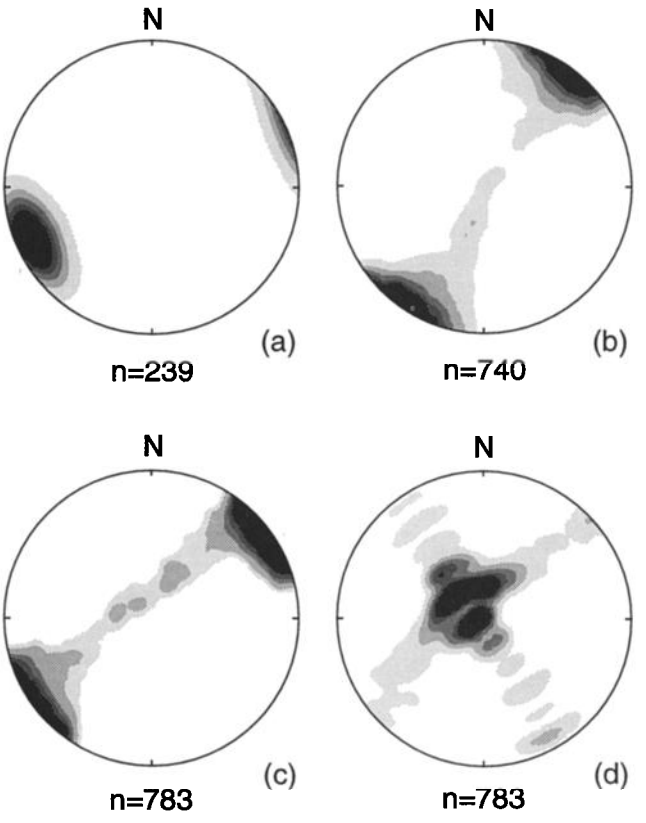


Figure 5. Contours of axes of principal stresses: (a) σ_1 axes, from site 11, contour increment is 10σ; (b) σ_1 axes, from site 23, contour increment is 6σ; (c) σ_1 axes, from site 19, contour increment is 6σ; (d) σ_2 axes, from site 19, contour increment is 4σ. Here n is the number of principal stresses.

each of the principal axes (Figure 5) which are probability distributions for the attitudes of the principal axes at each site. The maximum and minimum principal axes (σ_1 and σ_3) of the stress tensors for a single site possess distributions ranging from axial symmetry (e.g., Figure 5a), through skewed axial symmetry (e.g., Figure 5b), to girdle (e.g., Figure 5c), whereas the intermediate axes (σ_2) all exhibit a "cross" distribution (e.g., Figure 5d). To compare adjacent sites, we require statistical characterizations of these probability distributions, and in light of the generally nonaxial symmetry character of the distributions we have used Bingham's statistics [*Bingham*, 1974] rather than the common Fisher's statistics [*Fisher*, 1953] to calculate site means and their 95% confidence ellipses. The use of Bingham's statistics is described in the Appendix. The site-mean directions for σ_1, σ_2, and σ_3 are given in Figures 2 and 6a.

Determination of Rotational Axis and Angle

The site-mean directions of the σ_1 and σ_3 axes are all subhorizontal, whereas those of the σ_2 axes are subvertical (Figure 6a). Far from the FIF, different blocks (e.g., sites 22, 23, and 24) exhibit similar site-mean directions of principal stresses (N~33°E, Figures 2 and 6b), which are consistent with those expected in a stress field associated with the initiation of dextral transpressive displacement on the FIF. Approaching the fault, different sites within a single block (e.g., 14 and 18, 20 and 21; Figures 2 and 6b) show similar site-mean directions of principal stresses, but major changes occur between adjacent blocks. For example, the differences between sites 10 and 14, 18 and 20, and 21 and 22 are statistically significant at 95% confidence level because their 95% confidence limits do not overlap (Figure 6b). These major changes in the principal directions at the sites from near each other but from different blocks provide strong support for our assumption that the ranges of cleavage attitude reflect relative clockwise rotations between blocks rather than progressive variation in the cleavage attitudes.

In order to calculate the rotational axis relating the blocks to one another, the best-fit cone technique of *Ramsay* [1967] was applied to the site-mean σ_1, σ_2, and σ_3 axes (Figure 6a), resulting in three cone axes (Figure 6c). The *Fisher*'s [1953] mean of the cone axes was taken as the rotational axis. The resulting rotational axis plunges at 87.7° with azimuth 336.5° (the radius of 95% confidence circle α_{95} = 6.8°; Figure 6c). Using this rotational axis we calculated the rotational angle for each site with respect to the site-mean principal directions of site 22, which is one of the farthest sites from the FIF, by calculating dihedral angle between the plane common to the rotational axis and one of the site-mean principal axes of site 22 and the plane common to the rotational axis and the corresponding site-mean principal axis of the other site.

The resulting rotational angles show systematic but large variation over the area (Figure 7), reaching maximum values (58.7±3.3°) near the FIF and decreasing away from the fault (e.g., 0.0±1.6°, about 20 km distant). In keeping with the observations from the site means themselves, different sites within a single block yield similar rotational angles (e.g., sites 14 and 18, 20 and 21; Figures 2 and 7), but there is a significant difference between the rotations inferred for some sites from close to each other but from different blocks (e.g., sites 10 and 14, 18 and 20, 21 and 22; Figures 2 and 7). The mode of block rotations inferred in this area is similar to that described by *Nelson and Jones* [1987] in the Las Vegas Range and in contrast to the uniform rotation described and modelled elsewhere [*Geissman et al.*, 1989; *Ron et al.*, 1990]. The local block rotations (see also *MacConnell et al.*[1994]) may in part explain the apparent disparities between the paleomagnetic directions observed from the study area and North America. In the next section, we describe the paleomagnetic data from the area and discuss the effects of the block rotations on them.

Paleomagnetism of Plutonic Rocks

The simplest procedure for study of the effect of deformation on paleomagnetic data would have been to study the abundant bedded volcaniclastics of the Takla Group, since paleohorizontal can be determined. Pilot studies revealed, however, that their magnetization was not suitable, their unblocking temperatures and coercive forces being low. We had therefore to use plutonic rocks. These lacked any indication of paleohorizontal and this made the task more difficult.

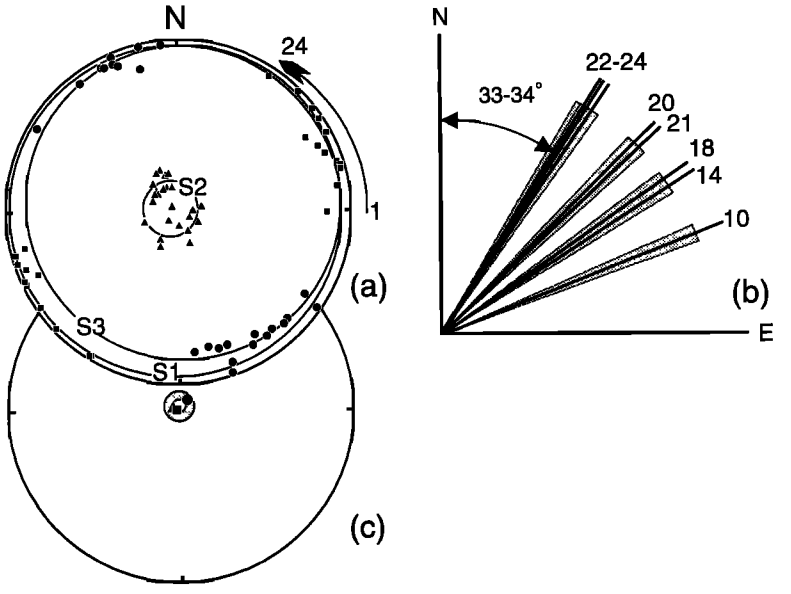


Figure 6. (a) Stereographic plot of site-mean principal stress axes. Solid squares, triangles, and circles are maximum (σ_1), intermediate (σ_2), and minimum (σ_3) principal stress axes, respectively; lines (S1, S2, and S3) are best-fitting small circles to σ_1, σ_2, and σ_3, respectively. (b) Plot of trends of σ_1 axes for some sites. Shaded areas are 95% confidence limits of the corresponding axes determined by projecting their 95% confidence ellipses, calculated using (A8) (see Appendix), onto the horizontal plane. Numbers show the sites. (c) Plot of cone axes. Solid square, triangle, and circle are cone axes of S1, S2, and S3 in Figure 6a, respectively; open circle is the mean rotational axis; shaded circle is 95% confidence circle (unless otherwise noted, the 95% confidence circle is determined using Fisher's statistics [*Fisher*, 1953]).

Sampling and Laboratory Procedure

The Early Jurassic to Cretaceous plutonic rocks [*Lord*, 1948; *Roots*, 1954; *Richards*, 1976a, b; *Woodsworth*, 1976] in the study area occur as dikes and stocks and range from fine-grained biotite granodiorite to coarse-grained quartz diorite. Samples were collected from 13 sites. Sites 3, 5, 6, and 7 are in the Cretaceous Johanson Creek stock [*Lord*, 1948; *Richards*, 1976a, b] in the Wrede Range (Figure 2), a fine-grained biotite granodiorite. Sites 9, 11, 19, and 20 are in the Early Jurassic coarse-grained quartz diorite Johanson Lake and Darb Lake stocks [*Lord*, 1948; *Richards*, 1976a, b] from which two hornblendes yield K-Ar ages of 174 ± 5 and 171 ± 5 Ma, respectively. Sites 15 and 17 are in the Cretaceous Kliyul Creek leuco-granodiorite body [*Lord*, 1948; *Richards*, 1976a, b; *Woodsworth*, 1976] from the southern extension of which a biotite yields a K-Ar age of 122 ± 6 Ma [*Wanless et al.*, 1972]. Sites 23, 26, and 27 are in the small dioritic dikes at Goldway Peak (Figure 2), which are probably coeval with the Cretaceous Johanson Lake stock to the north. At each site, two to three hand samples were collected as far away as possible from the faults. Samples were oriented by magnetic compass and checked by sun compass when possible.

In the laboratory, two to three cores were cut from each hand sample and two specimens from each core. A total of 77 cores (153 specimens) was obtained. Measurements were made on automated Schonstedt (SSM-1 and SSM-2) spinner magnetometers, from which four independent determinations of the magnetic vector were obtained. The precision of these four vectors (k_s) is a measure of within-specimen homogeneity. Here k_s was generally of the order of 10^2 to 10^3. Two specimens were chosen from each site for detailed stepwise demagnetization. One specimen was subjected to alternating field (AF) demagnetization using a Schonstedt GSD-5 tumbling demagnetizer; the other was thermally demagnetized in a Schonstedt TSD-1 furnace. According to the response of the two pilot specimens from each site to the thermal (up to 700°C) and AF (up to 100 mT) demagnetization, four or more steps of either thermal or AF cleaning treatment were applied to the rest, depending on how rapid the precision (k_s) dropped and whether an end point was achieved.

Description of Magnetization

Magnetization of the plutonic rocks varies and may be grouped into scattered magnetization, ill-defined magnetization, and well-clustered magnetization. Samples from sites 3, 9, 17, and 20 exhibit scattered magnetic directions, showing the typical characteristics of lightning-induced magnetization. Lightning is frequent in the mountain terrain. Within a single site, different samples, taken several metres apart, show scattered natural remanent magnetization (NRM) directions and unusually high NRM intensities (up to 38 A/m), and no coherent directions among the samples were obtained after either thermal or AF cleaning. During AF demagnetization by about 20 mT, the intensities of the specimens dropped to less than 10% of their NRM intensities; hence the magnetizations of the specimens are composed mainly of low

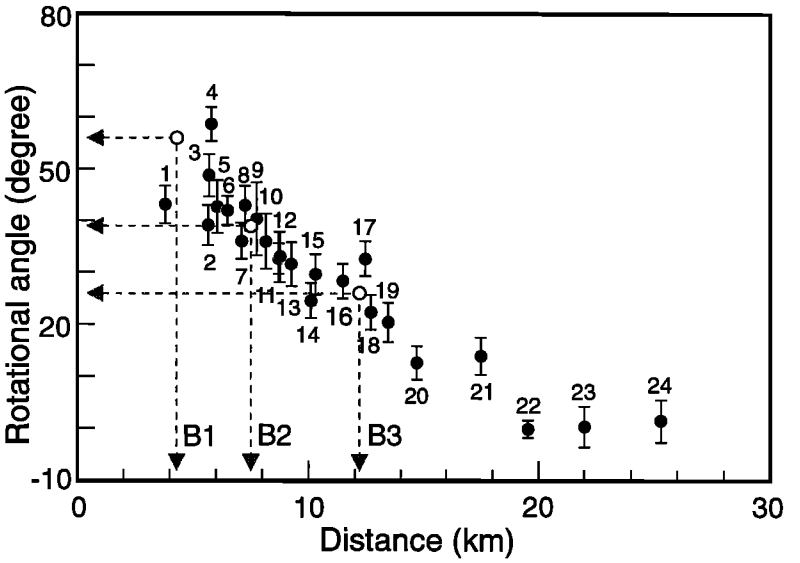


Figure 7. Plot of distance versus rotational angles [after *Zhang and Hynes*, 1994]. The abscissa represents the distance from the Finlay-Ingenika fault (FIF), and the ordinate represents the rotational angles. Bars are 95% confidence limits determined using the sum of long axes of the 95% confidence ellipses, calculated using (A8), for site 22 and the corresponding site. Open circles (B1, B2, and B3) are rotational angles for Blocks 1, 2, and 3, respectively.

coercivity components. They were not included in the subsequent analysis.

Sites 15, 23, and 26 are characterized by ill-defined magnetization, of which most specimens have low values of k_s (ranging from 1 to 100). During both thermal and AF cleaning, the magnetic directions of individual specimens changed rapidly and randomly, indicative of unstable magnetization, and after cleaning, no consistent magnetic directions within site were found. The magnetization at these sites is probably carried by coarse-grained magnetite as indicated by low coercivity (generally less than 20 mT). These sites were also excluded from the analysis.

Well-clustered magnetization was observed at 4 sites, 5, 6, 7, and 27. The NRM intensity at these sites ranges from 2.0 x 10^{-2} to 1.0 A/m (e.g., Figures 8-10). Sites 5 and 6 have high unblocking temperatures (570°-600°C) and at the cleaning step of 600°C the intensity of the specimens dropped to less than 15% of their NRM intensity, forming a shoulder characteristic of single-domain magnetite (e.g., Figure 8). AF demagnetization showed that a medium to high coercivity component (greater than 30 mT) is predominant, and after 10 mT the intensity decayed linearly according to the increase of demagnetization steps, forming a straight line on orthogonal plots (e.g., Figure 9). The magnetic directions between the cleaning steps of 200°-570°C and 20-80 mT are well-grouped, indicative of a stable magnetic component. In addition, microprobe analysis of the opaque minerals of the plutonic rocks indicates that they are mainly pure magnetite with minor ilmenite as exsolution lamellae within the magnetite, which is consistent with the observed unblocking temperature and coercivity. Therefore the main carriers of magnetization at these sites are probably fine-grained, single-domain magnetite grains.

The magnetization at site 7, however, is carried by both magnetite and hematite as indicated by a higher unblocking temperature and coercivity (greater than 600°C, e.g., Figure 10, and 100 mT). Since the hematite and magnetite have similar magnetic directions (e.g., the directions at 500°-600°C; Figure 10), they were both probably magnetized during the cooling of the magma. Since the specimen demagnetization trajectories were essentially linear on orthogonal intensity plots (e.g., Figures 9-10) the mean direction was determined by least squares line-fitting (LINEFIT) to the directions of removed components over the cleaning steps (e.g., 30-80 mT). All specimens from these sites show stable magnetization and reversed polarity (Figures 8-10) and are used in our analysis (Table 1).

An intermediate component was obtained using the removed magnetic vectors between 20 and 40 mT (Figure 11) from sites 11 and 19, the Early Jurassic Johanson Lake and Darb Lake stocks (with two hornblende K-Ar ages of 174±5 and 171±5 Ma, respectively). Because the inclinations of the magnetic directions (~70°, Figure 11) are much steeper than those observed from the adjacent Early Jurassic Hazelton Group rocks (47°-56°) [cf. *Monger and Irving*, 1980; *Vandall and Palmer*, 1990], we interpret magnetization of the intermediate component as a post-Jurassic overprint. The NRM directions of the specimens are scattered and centred about both present Earth's field (PEF) and the geocentric axial dipole field (GAD) (Figure 11). After cleaning, the removed magnetic directions become well-clustered and the PEF and GAD fields both lie outside the site mean's 95% confidence circles (Figure 11). Therefore the magnetization is neither a Bruhnes nor a present Earth field overprint. Since the polarity and directions of the magnetization are similar to those observed from the nearby mid-Cretaceous

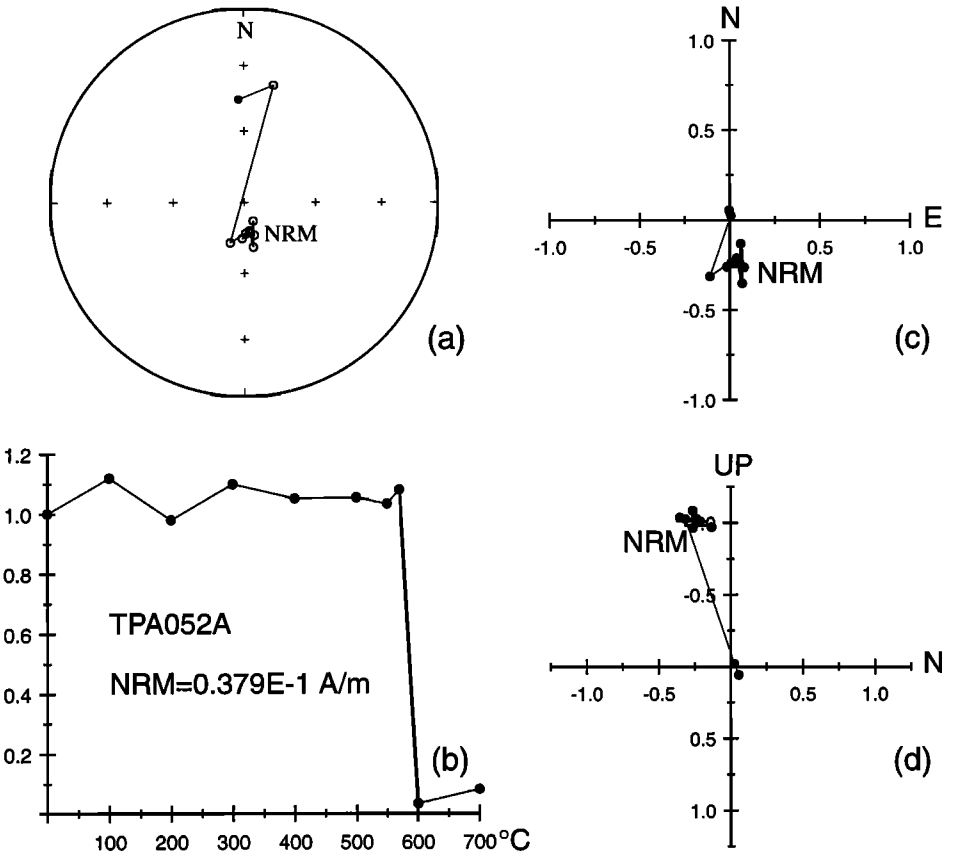


Figure 8. Thermal demagnetization for a representative specimen from site 5, showing changes in (a) orientation of the magnetic vector, (b) decay of the normalized magnetic intensity, and (c-d) decay of the normalized magnetic intensity on orthogonal plots, according to cleaning increments. NRM is natural remanent magnetization.

Axelgold [*Monger and Irving*, 1980] and Germansen [*Wynne et al.*, 1992] intrusions, it was probably acquired during the mid to Late Cretaceous. This is also supported by the frequent observations of magnetic overprint resulting from the mid Cretaceous plutonism near the study area [*Rees et al.*, 1985; *Butler et al.*, 1988]. The upper limit of the overprinting temperature was probably between 350°C and 525°C, since biotite K-Ar ages of Early Jurassic or older intrusions were reset to the mid to Late Cretaceous [*Rees et al.*, 1985, and references therein], while hornblende K-Ar ages like those we report, for which the unblocking temperature for Ar is ~525°C, were not. The magnetic directions of the intermediate components from sites 11 and 19 are included in our subsequent analysis (Table 1).

We were unable to separate the tiny biotite grains from our specimens of the Cretaceous Johanson Creek stock (sites 5, 6, and 7) and so obtained no radiometric date for it. Based on the polarity and magnetic directions, however, we believe the magnetization was acquired during the Late Cretaceous, after the Cretaceous Normal Superchron (83-124 Ma [cf. *Harland et al.*, 1989]), because it is dominated by reversed polarity and shows inclinations (~70°) comparable with those from the Late Cretaceous volcanics [*Marquis and Globerman*, 1988] to the north of the study area. We have therefore used the Late Cretaceous paleopole of cratonal North America (CNA) [cf. *Marquis and Globerman*, 1988] as a reference rather than the mid Cretaceous paleopole [cf. *Van Fossen and Kent*, 1992].

Isothermal Remanent Magnetization (IRM)

Low-field anisotropy of magnetic susceptibility (AMS) of specimens from site 5 was measured with an SI-2 magnetic susceptibility and anisotropy instrument (Sapphire Instruments, P.O. Box 385, Ruthven, Ontario, Canada). The results indicate that an AMS predominant foliation exists in the rocks, with $L = 1.07$ and $F = 1.37$ (L and F are magnetic lineation and foliation, respectively [cf. *Ellwood et al.*, 1988]). The magnetic foliation has an orientation similar to that of the rock foliation (striking 307° and dipping 69° northeast). Since the bulk susceptibility ($K_m = (K_1+K_2+K_3)/3 = 1.35 \times 10^{-3}$, where K_1, K_2, and K_3 are maximum, intermediate, and minimum magnetic susceptibility, respectively) and degree of anisotropy ($P = 1.45$) are not much greater than 10^{-3} and 1.35, respectively, the AMS is probably contributed by both ferromagnetic minerals (e.g., magnetite in our case) and rock matrix (e.g., biotite in our case) [cf. *Rochette*, 1987; *Borradaile*, 1988]. The question

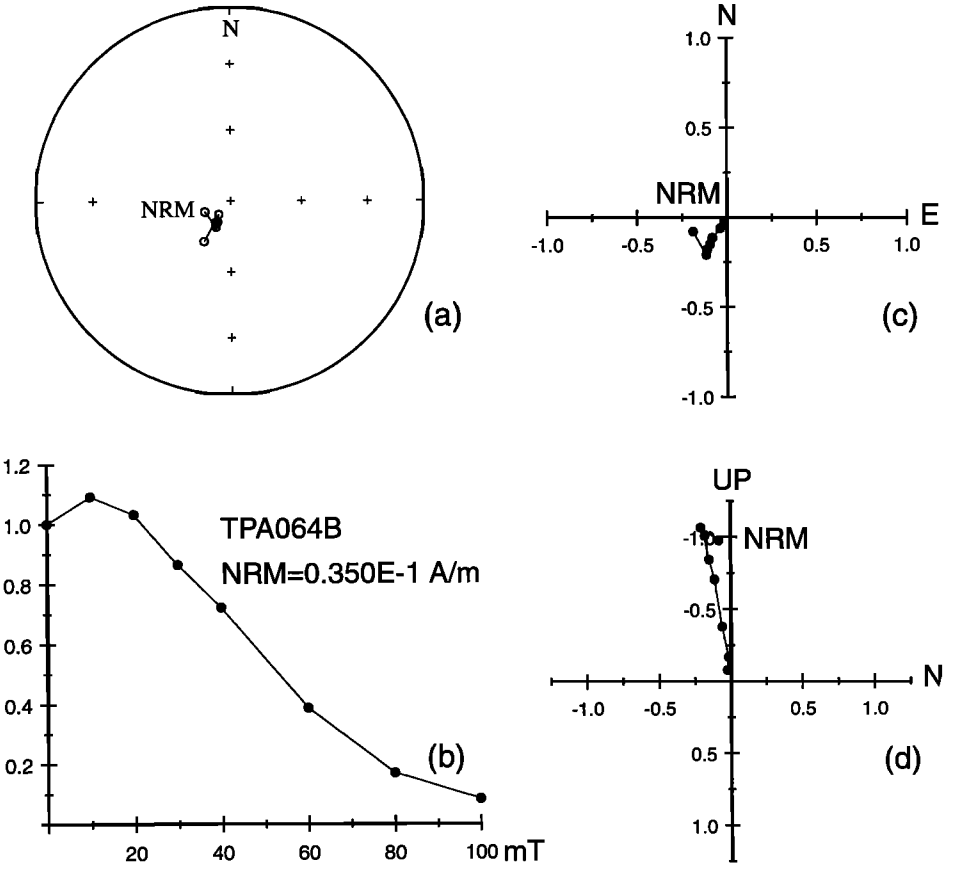


Figure 9. Alternating field (AF) demagnetization for a representative specimen from site 6. See legend for Figure 8.

is whether the AMS would affect the NRM directions of the specimens.

Because the characteristics of AF demagnetization of high-field (≥120 mT) IRM are similar to those of low-field (~5 mT) thermal remanent magnetization (TRM) for fine-grained single-domain (0.1 μm) magnetite dispersion, as demonstrated implicitly by *Lowrie and Fuller* [1971], we use a high-field IRM experiment to model the acquisition of low-field TRM which is probably the case for site 5. Two specimens were selected and magnetized in a DC field of 180 mT with a Sapphire Instrument SI-4 AF Demagnetizer after being demagnetized at 180 mT with the same instrument. Then each specimen was progressively cleaned by increments of 10 mT, up to 100 mT, using a Schonstedt GSD-5 tumbling demagnetizer. By repeating the above procedure, each specimen was magnetized three times in orthogonal directions (i.e., X, Y, and Z; see Figures 12a and 12b).

In all cases, the IRM directions depart significantly from the ambient field but are close to those theoretically predicted (Figures 12a and 12b), indicative of the effect of AMS on the IRM directions. During AF cleaning the magnetic directions moved toward the ambient field directions (Figures 12a and 12b), and after 20 or 30 mT the magnetic directions were well-clustered and essentially parallel to those of the ambient field (Figures 12a and 12b). The departures of mean directions [*Fisher*, 1953] at 20, or 30, to 100 mT from the ambient field directions were all within the range of 0.6° to 1.4° except for TPA054BY (Table 2 and Figure 12b), which is 4.4° and may reflect an error in orienting the specimen during IRM acquisition. This indicates that AMS has no effect on the direction of high coercivity (≥30 mT) stable components, which is consistent with the results for well-foliated gneissic rocks [cf. *Irving and Park*, 1973] and for laminated volcanic rocks [cf. *Marquis and Globerman*, 1988] and compatible with the experimental results of *Pozzi and Aïfa* [1989, and references therein]. The reason for this is presumably that the AMS is caused mainly by the alignment of coarse-grained, multidomain magnetite grains. Nevertheless, no corrections for the remanent magnetic directions are needed in our case because only the high coercivity components from these sites were used in the analysis.

Structural Interpretation of Paleomagnetic Data

The interpretable magnetic data (6 sites) are from three fault-bounded blocks: Block 1 (site 27), Block 2 (sites 5, 6, and 7), and Block 3 (sites 11 and 19) (see Table 1 and Figure 2). The paleopoles from the three blocks are scattered (Figure 13a), with a value of *Fisher*'s [1953] precision K = 48 (Table 1), but the blocks-mean paleopole is significantly different at 95% confidence level from the Late Cretaceous expected paleopole for CNA, as indicated by an F test (where

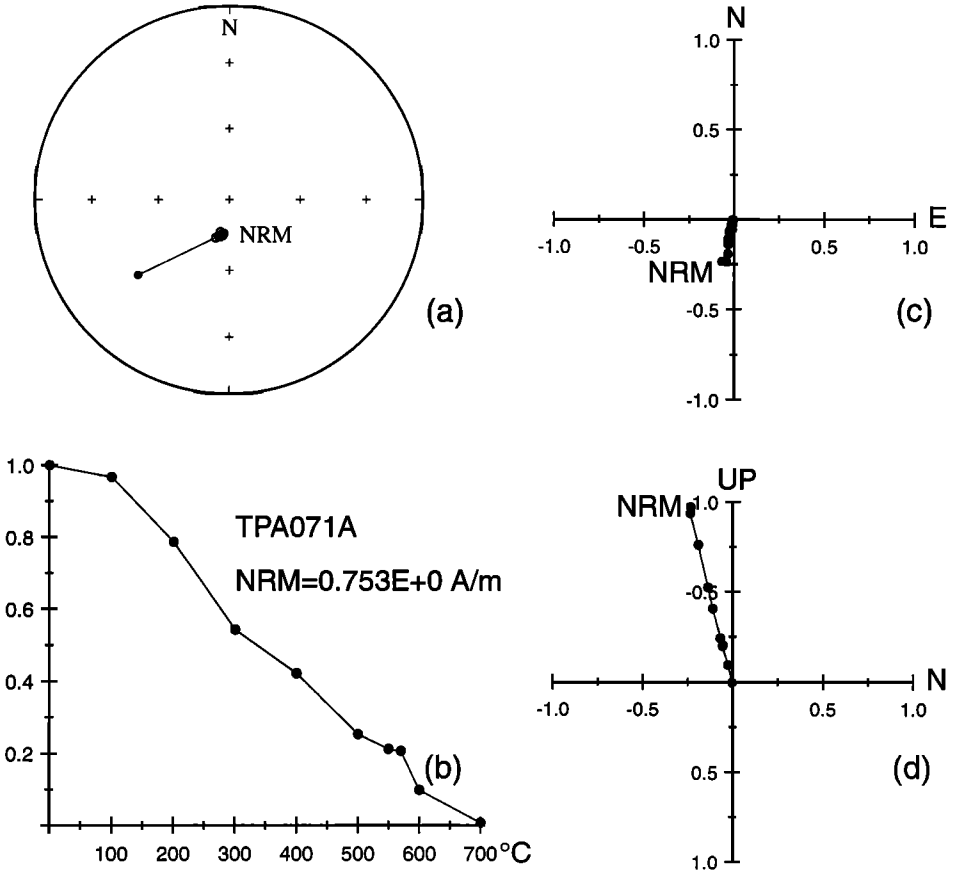


Figure 10. Thermal demagnetization for a representative specimen from site 7. See legend for Figure 8.

$F_{(2,8)} = 7.2 > F_{\alpha(2,8)}|_{\alpha = 0.05} = 4.46$ [cf. *Mardia*, 1972; *Upton and Fingleton*, 1989]), although their 95% confidence circles overlap (Figure 13a). Disparity between these two paleopoles shows an apparent northward paleolatitudinal displacement of $9 \pm 17°$ (Figure 14) and a clockwise rotation of $35 \pm 32°$ (Figure 15), which is compatible with other Cretaceous paleomagnetic data from the Intermontane Belt (Figures 14 and 15).

From the nearby cleavage data, we designated a rotational angle of 56° for Block 1, 39° for Block 2, and 26° for Block 3 (Figure 7). After unrotation about the axis determined using the cleavage data through these angles, the block-mean paleopoles moved closer to each other (Figure 13b), with a high value of Fisher's precision $K = 383$ (» $K = 48$ before corrections for the local block rotations, Table 1). The mean paleopole also moved closer to the expected paleopole (Figure 13b) but was still significantly different from it ($F_{(2,8)} = 6.0 > F_{\alpha(2,8)}|_{\alpha = 0.05} = 4.46$). The difference corresponds to an apparent paleolatitudinal displacement of $11 \pm 9°$ (Figure 14), but there is now no rotation ($-1 \pm 19°$, Figure 15). The local block rotations therefore account entirely for the anomalous magnetic declinations of the McConnell Creek region but not for the anomalous inclinations.

Geometrically, two hypotheses, tilting and northward transport, could explain the residual disparity between the expected and corrected paleopoles (Figure 13b). Tilting of ~7° to the north-northwest (N ~ 339°W) about a horizontal axis trending N ~ 069°E (see Table 1 for the expected and corrected magnetic directions) would remove the disparity. The required tilting axis is essentially perpendicular to the main trend of the structures in the area (Figure 2) and could not be achieved even if there were substantial dip-slip motion on many of the north-northwest trending faults within the region. There is no structural evidence for such a process. We therefore favour a process of northward transport. Northward transport of the region could have been accomplished by dextral transpressive motions on the Tintina trench and northern Rocky Mountain trench (TT-NRMT) fault zone [*Roddick*, 1967; *Tempelman-Kluit*, 1979; *Gabrielse*, 1985; *Struik*, 1993]. This process may be clearly illustrated by rotation about the Eulerian pole for the best-fitting small circle to the TT-NRMT fault zone [*Price and Carmichael*, 1986], which is located at latitude 35.4°N, longitude 201.3° E with an apical half angle of 31.2° (Figure 16a). Clockwise rotation about the Eulerian pole through 11.5° moves the corrected paleopole closer to the expected (Figures 16a and 16b). The corrected paleopole after rotation about the Eulerian pole is not significantly different from the expected ($F_{(2,8)} = 2.0 < F_{\alpha(2,8)}|_{\alpha = 0.05} = 4.46$). The resulting rotational angle of 11.5° along the small circle corresponds to a dextral displacement of ~670 km along the TT-NRMT fault zone, which is within the range from 450 to 900 km estimated based on the offsets of geological boundaries [*Roddick*, 1967; *Tempelman-Kluit*, 1979; *Gabrielse*, 1985].

Table 1. Paleomagnetic Data for Plutonic Rocks

Block	Site	Lat., °N, Long., °E	*n*, *s*	*D* deg	*I* deg	*k*	α_{95} deg	pole Lat., °N, Long., °E	*K*	A_{95} deg
1	TPA27	56.53, 233.75	6, 8	216	-71	394	3			
	BM	56.53, 233.75		216	-71			69.7, 312.2		
2	TPA05	56.63, 233.72	5, 10	186	-73	139	4			
	TPA06	56.63, 233.72	6, 12	196	-77	89	5			
	TPA07	56.63, 233.73	6, 11	190	-60	30	9			
	BM	56.63, 233.72		190	-70	83	14	83.9, 339.2		
3	TPA11	56.62, 233.85	6, 10	009	68	120	4			
	TPA19	56.52, 233.85	6, 10	350	67	70	4			
	BM	56.57, 233.85		359	67			83.7, 059.7		
BSM				194	-70	139	11	81.5, 332.7	48	18.0
BSMC				158	-69	1009	4	76.6, 134.0	383	6.3
K_{exp}				339	76		4	77.7, 185.8	255	7.7

Abbreviations are: lat. °N and long. °E, latitude north and longitude east; *n*, number of cores; *s*, number of specimens used for calculation of site mean; *D* and *I*, declination and inclination of the mean direction with respect to present horizontal; *k* (*K*), *Fisher*'s [1953] precision; α_{95}, half-angle of 95% confidence circle of the mean direction; A_{95}, half-angle of 95% confidence circle of blocks mean paleopole; BM, block mean; BSM, blocks mean; BSMC, blocks mean after corrections for the local block rotations; K_{exp}, Late Cretaceous reference magnetic direction or paleopole for cratonal North America (CNA) [*Marquis and Globerman*, 1988].

Discussion

Two major sources of uncertainty exist in our analysis. First, the number of sampling sites within individual blocks may not be enough to remove the paleomagnetic secular variation, and hence the error arising from the secular variation is not known. The block-mean paleopoles move closer after corrections for the local block rotations, indicating that the scatter of the block-mean paleopoles before corrections was probably mainly due to relative block rotations. This suggests that the error may not be significant. Second, uncertainty in the age of magnetization contributes some error. The choice of the mid Cretaceous reference paleopole [*Van Fossen and Kent*, 1992] would affect the fitness of our results. It would, however, not change our estimate of the displacement significantly as this reference paleopole is only slightly different from the one we used [*Marquis and Globerman*, 1988]. For example, the estimated northward displacement by using the mid Cretaceous reference pole (71.2°N, 194.1°E, A_{95} = 3.7° [cf. *Van Fossen*

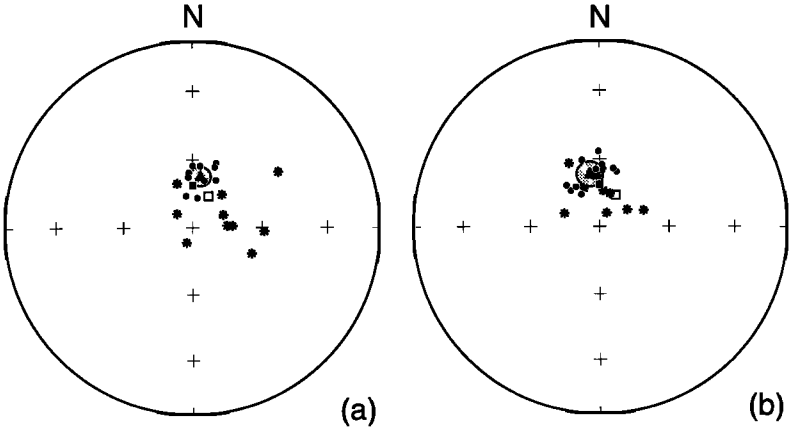


Figure 11. Stereographic plots of magnetic directions of sites (a) 11 and (b) 19. Stars are NRM directions; solid circles are magnetic directions removed at 20-40 mT; open square is the present Earth field; solid square is the geocentric axial dipole field; solid triangle is the site mean of the removed directions; shaded circle is the 95% confidence circle.

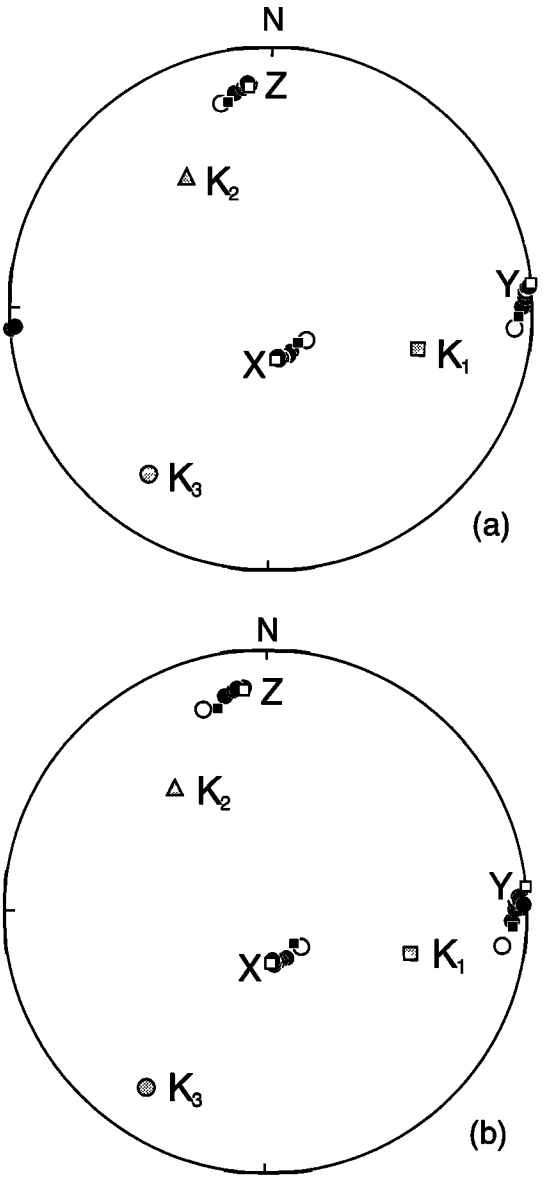


Figure 12. Plots showing changes of the isothermal remanent magnetization (IRM) directions of specimens (a) TPA052B and (b)TPA054B, respectively, during AF cleaning. Shaded square, triangle, and circle are the directions of K_1, K_2 and K_3; solid squares are predicted directions, calculated by the linear relationship $\mathbf{m} = \mathbf{Kh}$, where **K** is the second-rank tensor of magnetic susceptibility and **h** is a column vector specifying the ambient field direction; open squares are ambient field directions; open circles are IRM directions; solid circles are magnetic directions at 10-100 mT.

and Kent, 1992]) is ~740 km, but the angular distance between the observed and the reference poles is 14.5°, that is, greater than 10.0° (the sum of their 95% confidence limits, see Table 1 for A_{95} of the observed pole). Nevertheless, the amount of northward displacement estimated from our study is in good agreement with that determined from the offsets of geological boundaries along the TT-NRMT fault zone.

Our study demonstrates again that to understand the processes responsible for the observed discordant paleomagnetic directions in complexly deformed regions, the most important problem is the determination of paleohorizontal at the time when the magnetization was acquired. Barometric analysis of the contact metamorphic assemblage around a pluton could provide an estimate of postemplacement tilting (rotation about a horizontal axis) of the paleohorizontal [e.g., *Irving and Archibald*, 1990; *Brown and Burmester*, 1991; *Ague and Brandon*, 1992], whereas structural analysis could constrain rotations of the paleohorizontal about either horizontal or vertical axes, as demonstrated from our study. It would therefore be extremely beneficial if paleomagnetic studies could be accompanied with detailed petrological and structural studies in regions in which rocks experienced a complex deformation history.

Other Cretaceous paleomagnetic data from the Intermontane Belt show apparent northward paleolatitudinal displacement of 9.5°-23° (Figure 14) and clockwise rotations ranging from 9° to 75° (Figure 15). These inconsistencies in the observed discordant paleomagnetic directions require that the Intermontane Belt behaved as different discrete blocks rather than a coherent composite terrain during the Late

Table 2. Results of Isothermal Remanent Magnetization (IRM) Experiments

Specimen	Direction	D, deg	I, deg	k	α_{95}, deg	def, deg
TPA052BX	PD	140.9	76.4			
	IRM	131.2	75.3			
	Field	174.0	74.0			
	30-100 mT	169.0	74.4	16009	0.53	1.4
TPA052BY	PD	091.5	5.7			
	IRM	094.1	7.2			
	Field	084.0	0.0			
	30-100 mT	085.1	0.8	2255	1.41	1.4
TPA052BZ	PD	347.9	20.6			
	IRM	345.9	20.6			
	Field	354.0	16.0			
	20-100 mT	353.8	15.4	9046	0.63	0.6
TPA054BX	PD	137.4	76.8			
	IRM	133.6	74.5			
	Field	174.0	74.0			
	30-100 mT	170.0	73.9	14423	0.56	1.1
TPA054BY	PD	093.0	5.8			
	IRM	097.7	9.2			
	Field	084.0	0.0			
	30-100 mT	086.8	3.4	3901	1.07	4.4
TPA054BZ	PD	346.2	21.6			
	IRM	342.3	20.5			
	Field	354.0	16.0			
	20-100 mT	353.1	15.4	5627	0.80	1.1

Abbreviations are: PD, predicted direction (see legend for Figure 12); IRM, direction before cleaning; Field, ambient field direction; 30-100 mT, *Fisher*'s [1953] mean of the directions between 30 and 100mT; D and I, declination and inclination; k, Fisher's precision; α_{95}, half-angle of 95% confidence circle; def, angle between ambient field and high coercivity component directions.

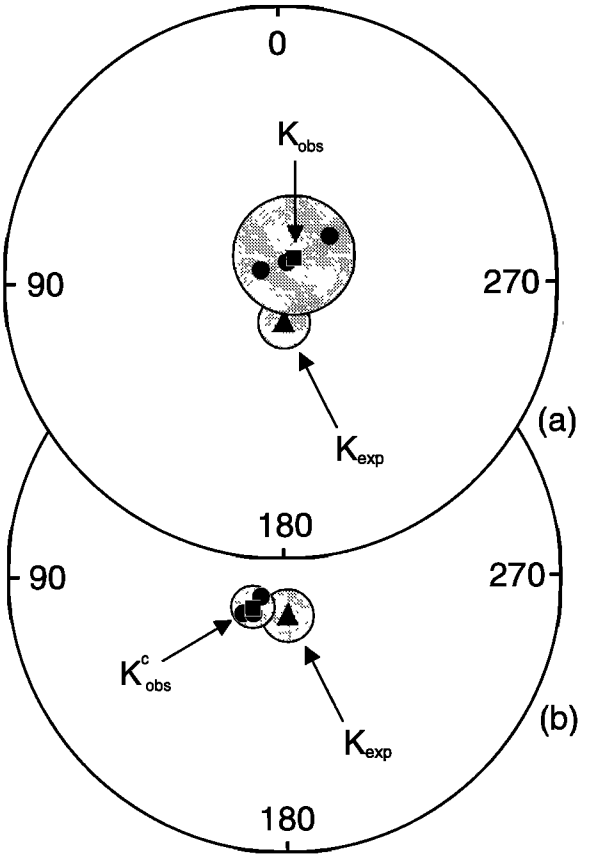


Figure 13. Stereographic plots of paleomagnetic poles showing observed paleopoles (a) before and (b) after corrections for the local block rotations. Solid triangle (K_{exp}) is expected Late Cretaceous paleopole for cratonal North America (CNA) (Table 1); solid circles are observed paleopoles from three fault-bounded blocks; solid square is blocks mean (K_{obs} and K^c_{obs} are the paleopoles before and after the corrections, respectively); shaded circles are 95% confidence circles.

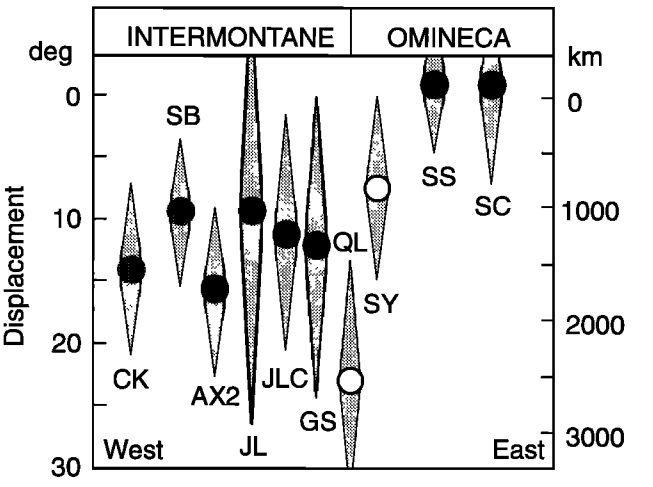


Figure 14. Plot of Cretaceous paleomagnetic data observed from the Intermontane and Omineca Belts showing apparent northward displacements. Solid circles are intrusives; shaded circles are bedded rocks; open circles are overprint; shaded areas are 95% confidence limits. References are SC and SS, *Irving and Archibald* [1990]; SY, *Butler et al.* [1988]; QL, *Rees et al.* [1985]; GS, *Wynne et al.* [1992]; AX2, *Monger and Irving* [1980]; SB, *Irving et al.* [1995]; CK, *Marquis and Globerman* [1988]; JL and JLC, this study (before and after corrections for the local block rotations, respectively).

Cretaceous to Eocene deformation associated with the oblique convergence of the Kula plate against North America. The estimated northward displacement of ~670 km from our study is less than the recent estimate of 1100±600 km from the well-constrained Spences Bridge Group volcanics in the west [*Irving et al.*, 1995], indicating that there could be an increase in apparent northward paleolatitudinal displacement from east to west across the Intermontane Belt. This westward increase of northward displacement suggests that the dextral strike-slip motions may not only have occurred on the eastern boundary of the Intermontane Belt, the TT-NRMT fault zone, but also have been distributed within the belt. Compared with the estimates obtained from the Coast Belt (of the order of 3000 km [cf. *Irving and Wynne*, 1990; *Wynne et al.*, 1995, and references therein]), however, the northward displacement of the Intermontane Belt estimated from our study and another recent study [*Irving et al.*, 1995] is much lesser. Our results therefore do not disprove the northward transport model of the western allochthonous terrains but suggest that the major tectonic break is not located at the boundary between the Intermontane and Omineca belts but lies somewhere to the west of the Intermontane Belt as proposed by *Irving et al.* [1995].

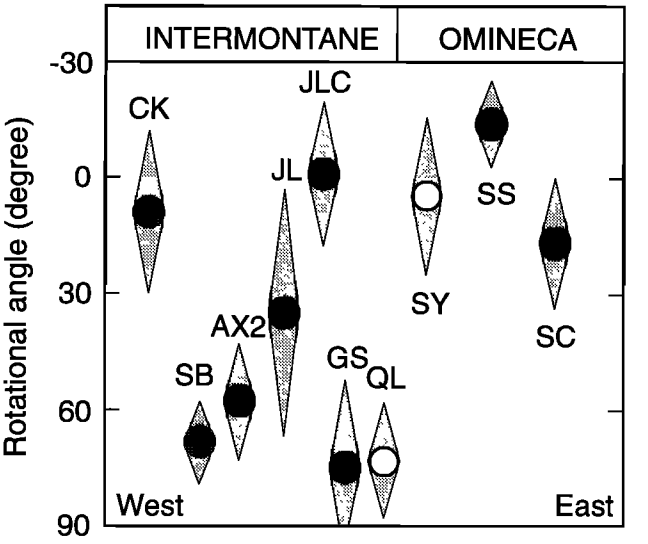


Figure 15. Plot of Cretaceous paleomagnetic data observed from the Intermontane and Omineca Belts showing apparent clockwise (positive) rotations. See legend for Figure 14.

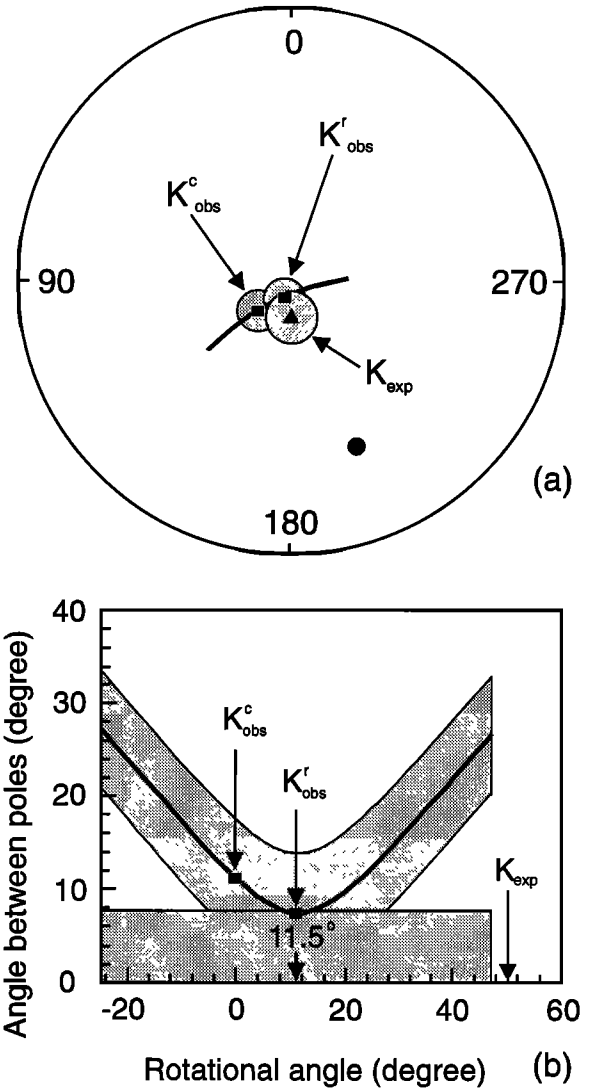


Figure 16. Plots showing clockwise rotation of the observed paleopole (K^c_{obs}) about the Eulerian pole of *Price and Carmichael* [1986]. Solid circle is the Eulerian pole; K^r_{obs} is the paleopole after rotation about the Eulerian pole; other symbols are as for Figure 13.

Appendix

Bingham's statistics [*Bingham*, 1974] describe an antipodally symmetric distribution of directional data on a unit sphere. Given $\mathbf{L}_1, \mathbf{L}_2, \ldots, \mathbf{L}_n$ ($\mathbf{L}_i = (l_i, m_i, n_i)^T$, unit column vectors representing the directions of axial data), a random sample from a population with a Bingham's distribution, its probability density function may be written

$$f(\mathbf{L}, \mathbf{k}, \mathbf{u}) = \frac{1}{4\pi d(\mathbf{k})} \exp\ [tr(\mathbf{k}\mathbf{u}^T\mathbf{L}\mathbf{L}^T\mathbf{u})]$$

$$= \frac{1}{4\pi d(\mathbf{k})} \exp\ [k_1(\mathbf{u}_1{}^T\mathbf{L})^2 + k_2(\mathbf{u}_2{}^T\mathbf{L})^2 + k_3(\mathbf{u}_3{}^T\mathbf{L})^2] \quad \text{(A1)}$$

where $\mathbf{u} = (\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$ is an orthogonal matrix specifying three true orthogonal (normalized) directions, $\mathbf{k} = \mathrm{diag}(k_1, k_2, k_3)$ is a diagonal matrix of precision coefficients for the distribution, $d(\mathbf{k})$ is a constant given by

$$\frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi} \exp(k_1 \cos^2\phi \sin^2\theta + k_2 \sin^2\phi \sin^2\theta + k_3 \cos^2\theta) \sin\theta\, d\theta d\phi$$

(where $\sin\theta\cos\phi = \mathbf{u}_1{}^T\mathbf{L}$, $\sin\theta\sin\phi = \mathbf{u}_2{}^T\mathbf{L}$, and $\cos\theta = \mathbf{u}_3{}^T\mathbf{L}$), $tr()$ represents the sum of the diagonal elements of matrix inside the parentheses, and superscript (T) represents the transpose of its corresponding matrix or vector (c. Bingham, 1964, as referenced by *Bingham* [1974])[*Mardia*, 1972; *Upton and Fingleton*, 1989].

In terms of classical statistics, the three mean orthogonal directions ($\hat{\mathbf{u}}$) of an observed sample can be obtained from the maximum likelihood estimate (m.l.e.) of $\mathbf{u}$ [*Rice*, 1988]. The likelihood function [*Rice*, 1988] of $\mathbf{k}$ and $\mathbf{u}$ in (A1) is found to be

$$L(\mathbf{L}_1, \mathbf{L}_2, \ldots, \mathbf{L}_n, \mathbf{k}, \mathbf{u})$$

$$= \prod_{i=1}^{n} \frac{1}{4\pi d(\mathbf{k})} \exp\ [tr(\mathbf{k}\mathbf{u}^T\mathbf{L}_i\mathbf{L}_i{}^T\mathbf{u})]$$

$$= \left(\frac{1}{4\pi d(\mathbf{k})}\right)^n \exp\ [tr(\mathbf{k}\mathbf{u}^T\mathbf{T}\mathbf{u})] \quad \text{(A2)}$$

where

$$\boldsymbol{T} = \begin{pmatrix} \sum_{i=1}^{n} l_i^2 & \sum_{i=1}^{n} l_i m_i & \sum_{i=1}^{n} l_i n_i \\ & \sum_{i=1}^{n} m_i^2 & \sum_{i=1}^{n} m_i n_i \\ & & \sum_{i=1}^{n} n_i^2 \end{pmatrix}$$

is a symmetric matrix. Its logarithm may be written

$$\ln(L) = -n\ln(4\pi) - n\ln(d(\mathbf{k})) + tr(\mathbf{k}\mathbf{u}^T\mathbf{T}\mathbf{u}) + tr(\mathbf{A}(\mathbf{I} - \mathbf{u}^T\mathbf{u})) \quad \text{(A3)}$$

where $\mathbf{u}^T\mathbf{u} = \mathbf{I}$, because $\mathbf{u}$ is an orthogonal matrix (equation (A1)) and $\mathbf{A}$ is a symmetric matrix of Lagrangian multipliers [*Mardia*, 1972]. Based on statistical estimate theory, the sample $\hat{\mathbf{u}}$ directions can be obtained when $\ln(L)$ of (A3) achieves maximum value [*Rice*, 1988]. Thus

$$\frac{\partial(\ln(L))}{\partial \mathbf{u}} = tr(\mathbf{k}\mathbf{u}^T\mathbf{T}) + tr(-\mathbf{A}\mathbf{u}^T) = 0 \quad \text{(A4)}$$

By rewriting equation (A4) we have

$$\mathbf{k}^{-1}\mathbf{A} = \mathbf{k}^{-1}\mathbf{k}\hat{\mathbf{u}}^T\mathbf{T}\hat{\mathbf{u}} = \hat{\mathbf{u}}^T\mathbf{T}\hat{\mathbf{u}} \quad \text{(A5)}$$

where $\mathbf{k}^{-1}$ is the inverse of $\mathbf{k}$ and $\mathbf{A}$ is a diagonal matrix as

required by the right side of the equation. It is evident from (A5) that when (A4) reaches its maxima, $\mathbf{k}^{-1}\mathbf{A} = \mathrm{diag}(\tau_1, \tau_2, \tau_3)$ and $\hat{\mathbf{u}} = (\mathbf{t}_1, \mathbf{t}_2, \mathbf{t}_3)$, where (τ_1, τ_2, τ_3) and $(\mathbf{t}_1, \mathbf{t}_2, \mathbf{t}_3)$ are the minimum, intermediate, and maximum eigenvalues and eigenvectors of matrix $\mathbf{T}$, respectively. Therefore the m.l.e. of $\mathbf{u}$ is the eigenvectors of $\mathbf{T}$.

The geometrical significance of the eigenvalues could be clearly illustrated by the analysis of moment of inertia (i.e., the sum of squares of the perpendicular distances of the observed sample (l_i, m_i, n_i) from the directions $\hat{\mathbf{u}}$ [cf. *Mardia*, 1972]). For example, when $\tau_1 \approx \tau_2 \approx \tau_3$, the distribution is uniform, when $\tau_1 \approx \tau_2 < \tau_3$, it is axially symmetric (either unipolar or bipolar) about $\mathbf{t}_3$, and when $\tau_1 < \tau_2 \approx \tau_3$, it is a symmetric girdle distribution in the plane normal to the axis $\mathbf{t}_1$, etc.

The sample precision coefficients $\hat{\mathbf{k}}$ can similarly be obtained from the m.l.e of $\mathbf{k}$. The sum of k_i (i = 1 to 3) of (A1) is arbitrary, so that k_3 may be set to zero [cf. *Mardia*, 1972; *Bingham*, 1974]. Substituting $\hat{\mathbf{u}}$ into (A3) leads to

$$\ln(L) = -n\ln(4\pi) - n\ln(d(\mathbf{k})) + k_1\tau_1 + k_2\tau_2 \qquad \text{(A6)}$$

Maximization of $\ln(L)$ in (A6) as a function of k_1 and k_2 yields two nonlinear simultaneous equations, from which $\hat{\mathbf{k}}$ could be calculated. Here $\hat{\mathbf{k}}$ may also be solved, however, by calculating the maxima of (A6) by changing the values of k_1 and k_2, so that

$$\ln(L) = -n\ln(4\pi) - n\ln(d(k_1, k_2)) + k_1\tau_1 + k_2\tau_2$$

$$= \ln(L)_{\max}|_{k_1 = \hat{k}_1,\ k_2 = \hat{k}_2} \qquad \text{(A7)}$$

Equation (A7) may be solved numerically by the method of *Fletcher and Powell* [1963]. This is the procedure we adopted. Finally, based on a chi-square test, estimates of the axes of the 95% confidence ellipses about the $\hat{\mathbf{u}} = (\mathbf{t}_1, \mathbf{t}_2, \mathbf{t}_3)$ directions are found to be

$$\hat{\alpha}_{ij} = \left(\frac{\chi^2_{1-\alpha}(2)}{2(\tau_i - \tau_j)(\hat{k}_i - \hat{k}_j)}\right)^{1/2} \qquad \text{(A8)}$$

in the directions of $\mathbf{t}_i$-$\mathbf{t}_j$ on a unit sphere [cf. *Bingham*, 1974], where $i, j = 1, 3$ and $i \neq j$.

The maximum eigenvectors ($\mathbf{t}_3$) were taken as site-mean directions for the principal axes (σ_1, σ_2, and σ_3) of the stress tensors, with 95% confidence limits of $\hat{\alpha}_{31}$ and $\hat{\alpha}_{32}$. A Fortran program was used to calculate the eigenvectors $\mathbf{t}_1$, $\mathbf{t}_2$, and $\mathbf{t}_3$ (equation (A5)); precision coefficients $\hat{k}_1$ and $\hat{k}_2$ (equation (A7)); and 95% confidence limits $\hat{\alpha}_{ij}$ (equation (A8)) of the stress tensors.

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